Assignment: Chronic Kidney Disease

Introduction

The given dataset given by the hospital has 399 patients records under 25 columns. Using this dataset, we need to build a model that should predict a upcoming patient who has disease or not.

There are 249 records classified as patients and 150 are not-patients. Since 62% of the data classified as patients and 38% of data classified as not-patients. So, this dataset is imbalanced.

Also, the dataset has categorical data, we went through pre-processing by encoding.

Lab

Please find the notebook in the same repo.

Model Evaluation

Went through following classifiers, and below table shows the summary of confusion matrix and classification report. Accuracy, precision and re-call values nearly same for all algorithms and when compared to Type-1 and Type-2 errors the Logistic Regression wins.

Algorithm	T-Positive	T-Negative	TYPE-1	TYPE-2	Accuracy	TYPE Comparison	
Random Forest Classifier	44	74	1	1	0.99	Type 1 = Type 2	Precision/Recall ratio is good
Logistic Regression (CV)	45	74	0	1	0.99	Type 1 < Type 2	Precision/Recall ratio is good
SVM Classifier (CV)	44	71	1	4	0.96	Type 1 < Type 2	Accuracy is less than others
Decision Tree Classifier (CV)	45	72	0	3	0.97	Type 1 < Type 2	Accuracy is less than others

Detailed classification report.

1. Random Forrest Classifier

```
              precision    recall  f1-score   support

           0       0.98      1.00      0.99        45
           1       1.00      0.99      0.99        75

    accuracy                           0.99       120
   macro avg       0.99      0.99      0.99       120
weighted avg       0.99      0.99      0.99       120
```

2. Logistic Regression (CV)

Classification – Assignment

	precision	recall	f1-score	support
0	1.00	1.00	1.00	45
1	1.00	1.00	1.00	75
accuracy			1.00	120
macro avg	1.00	1.00	1.00	120
weighted avg	1.00	1.00	1.00	120

3. SVM Classifier (CV)

	precision	recall	f1-score	support
0	0.92	0.98	0.95	45
1	0.99	0.95	0.97	75
accuracy			0.96	120
macro avg	0.95	0.96	0.96	120
weighted avg	0.96	0.96	0.96	120

4. Decision Tree Classifier (CV)

	precision	recall	f1-score	support
0	0.94	1.00	0.97	45
1	1.00	0.96	0.98	75
accuracy			0.97	120
macro avg	0.97	0.98	0.97	120
weighted avg	0.98	0.97	0.98	120